

Topology

Lecture 05: Homeomorphism

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1 Introduction

In topology we study properties of spaces that are preserved under continuous deformation — stretching, bending, twisting — but *not* tearing or gluing. A **homeomorphism** formalises exactly this idea: two spaces are *topologically the same* if there is a bicontinuous bijection between them.

[!] Intuition: Clay Analogy

Imagine two objects made of infinitely flexible clay. If you can deform one into the other without ever tearing the clay or gluing separate pieces together, the two shapes are **homeomorphic**. The key operations that *destroy* homeomorphism are:

- **Tearing** — points that were close suddenly map far apart (continuity fails).
- **Gluing** — two distinct points collapse to one (bijectivity fails).

2 Formal Definition

[Def] Homeomorphism

Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function $\varphi : X \rightarrow Y$ is a **homeomorphism** if it satisfies all three conditions:

- (i) φ is **bijective** (one-to-one and onto), so $\varphi^{-1} : Y \rightarrow X$ exists.
- (ii) φ is **continuous**: every open set in Y has an open pre-image in X .
- (iii) φ^{-1} is **continuous**: every open set in X has an open pre-image in Y .

A function satisfying (ii) and (iii) is called **bicontinuous**. When a homeomorphism exists we say the spaces are **homeomorphic** and write $X \cong Y$.

The key identities that make φ and φ^{-1} inverses are:

$$\varphi^{-1} \circ \varphi = \text{id}_X \quad \text{and} \quad \varphi \circ \varphi^{-1} = \text{id}_Y.$$

[!] Why Not Just Bijective + Continuous?

For groups and vector spaces a bijective homomorphism automatically has a structure-preserving inverse. For topological spaces this *fails*: there exist bijective continuous functions whose inverse is discontinuous. We must therefore check continuity of φ^{-1} separately.

2.1 What Goes Wrong Without All Three Conditions

Violation	Effect	Condition that fails
Tearing the space	Close points map far apart	φ not continuous
Gluing two points	Two inputs share one output	φ not injective
Inverse tears	Going back rips the space	φ^{-1} not continuous

3 Homeomorphic is an Equivalence Relation

[Thm] Equivalence Relation

Being homeomorphic is an equivalence relation on any collection of topological spaces.

Property	Why it holds
Reflexive: $X \cong X$	id_X is a homeomorphism.
Symmetric: $X \cong Y \Rightarrow Y \cong X$	φ^{-1} is itself a homeomorphism.
Transitive: $X \cong Y, Y \cong Z \Rightarrow X \cong Z$	Compose the two homeomorphisms.

4 Key Theorem — The Open-Set Criterion

[Thm] Open-Set Criterion (★)

A bijection $f : (X_1, T_1) \rightarrow (X_2, T_2)$ is a homeomorphism **if and only if**

$$U \in T_1 \iff f(U) \in T_2. \quad (\star)$$

That is, f sets up a one-to-one correspondence between the open sets of X_1 and the open sets of X_2 .

4.1 Proof of the Open-Set Criterion

Direction 1: Assume f is a homeomorphism. We show (★).

Step 1. Take any $U \in T_1$. We want to show $f(U) \in T_2$.

Step 2. Key identity: $f(U) = (f^{-1})^{-1}(U)$, i.e. $f(U)$ is the pre-image of U under f^{-1} .

Step 3. Since f is a homeomorphism, f^{-1} is continuous, so the pre-image of any open set under f^{-1} is open. Since U is open, $f(U)$ is open. ✓

Step 4. Conversely, if $f(U) \in T_2$, then $U = f^{-1}(f(U))$ and f is continuous, so $U \in T_1$. ✓

Direction 2: Assume (★) holds. We show f is a homeomorphism.

Step 1. f is continuous. Take $V \in T_2$. Write $V = f(f^{-1}(V))$. Set $u = f^{-1}(V)$. By (★), since $f(u) = V \in T_2$, we get $u \in T_1$. So $f^{-1}(V)$ is open. ✓

Step 2. f^{-1} is continuous. Take $U \in T_1$. The pre-image of U under f^{-1} is $f(U)$. By (★), $f(U) \in T_2$ (open). ✓

5 Worked Examples

5.1 Example 1 — Finite Topological Spaces

[Ex] Example 1: Finite spaces (direct use of $(\star)$)

Let $X = \{1, 2, 3\}$ with $T_1 = \{\emptyset, \{1\}, \{1, 2\}, X\}$ and $Y = \{a, b, c\}$ with $T_2 = \{\emptyset, \{a\}, \{a, b\}, Y\}$. Define $f : X \rightarrow Y$ by $f(1) = a$, $f(2) = b$, $f(3) = c$.

Step 1. Bijective. Every element of Y is the image of exactly one element of X . ✓

Step 2. Verify $(\star)$: check each open set $U \in T_1$.

$U \in T_1$	$f(U)$	$f(U) \in T_2?$
$\emptyset$	$\emptyset$	Yes ✓
$\{1\}$	$\{a\}$	Yes ✓
$\{1, 2\}$	$\{a, b\}$	Yes ✓
X	Y	Yes ✓

Step 3. Verify $(\star)$: check each open set $V \in T_2$.

$V \in T_2$	$f^{-1}(V)$	$f^{-1}(V) \in T_1?$
$\emptyset$	$\emptyset$	Yes ✓
$\{a\}$	$\{1\}$	Yes ✓
$\{a, b\}$	$\{1, 2\}$	Yes ✓
Y	X	Yes ✓

All open sets match perfectly. By $(\star)$, f is a homeomorphism, so $X \cong Y$.

5.2 Example 2 — Linear Homeomorphism on $\mathbb{R}$

[Ex] Example 2: Show $f(x) = 2x + 3$ is a homeomorphism from $\mathbb{R}$ to $\mathbb{R}$

Step 1. Bijective.

- *Injective:* If $f(x_1) = f(x_2)$, then $2x_1 + 3 = 2x_2 + 3$, so $x_1 = x_2$. ✓
- *Surjective:* Given $y \in \mathbb{R}$, set $x = (y - 3)/2 \in \mathbb{R}$. Then $f(x) = y$. ✓

Step 2. f is continuous. $f(x) = 2x + 3$ is a polynomial, hence continuous on $\mathbb{R}$. ✓

Step 3. Find f^{-1} . Solving $y = 2x + 3$ for x :

$$f^{-1}(y) = \frac{y - 3}{2}.$$

Step 4. f^{-1} is continuous. $f^{-1}(y) = \frac{y-3}{2}$ is also a polynomial (linear), hence continuous. ✓

Step 5. Verify. $f^{-1}(f(x)) = \frac{(2x+3)-3}{2} = x$. $f(f^{-1}(y)) = 2 \cdot \frac{y-3}{2} + 3 = y$. ✓

Conclusion: f is a homeomorphism, so $\mathbb{R} \cong \mathbb{R}$ (via this linear map).

5.3 Example 3 — Any Two Open Intervals are Homeomorphic

[Ex] Example 3: Show $(0, 1) \cong (2, 5)$

We need a continuous bijection that maps $0 \mapsto 2$ and $1 \mapsto 5$.

Step 1. Construct φ . A linear map sending $0 \rightarrow 2$ and $1 \rightarrow 5$ has slope $(5-2)/(1-0) = 3$ and intercept 2:

$$\varphi(x) = 2 + 3x, \quad x \in (0, 1).$$

Step 2. Check endpoints: $\varphi(0) = 2$ and $\varphi(1) = 5$, so φ maps $(0, 1)$ into $(2, 5)$. ✓

Step 3. Bijective. φ is strictly increasing (slope $3 > 0$), hence injective. Given $y \in (2, 5)$, set $x = (y - 2)/3 \in (0, 1)$. Surjective. ✓

Step 4. Continuous. $\varphi(x) = 2 + 3x$ is a polynomial. ✓

Step 5. Inverse:

$$\varphi^{-1}(y) = \frac{y - 2}{3}, \quad y \in (2, 5).$$

Verify: $\varphi^{-1}(\varphi(x)) = \frac{(2 + 3x) - 2}{3} = x$. ✓

Step 6. φ^{-1} is also a polynomial, hence continuous. ✓

Conclusion: $(0, 1) \cong (2, 5)$. More generally, any two open intervals in $\mathbb{R}$ are homeomorphic via a suitable linear map.

5.4 Example 4 — Open Unit Ball Homeomorphic to all of $\mathbb{R}^n$

[Ex] Example 4: Show $(0, 1) \cong \mathbb{R}$ (bounded $\cong$ unbounded)

Step 1. The homeomorphism. Define

$$f : (0, 1) \rightarrow \mathbb{R}, \quad f(x) = \tan\left(\pi\left(x - \frac{1}{2}\right)\right).$$

Step 2. Range covers $\mathbb{R}$. As $x \rightarrow 0^+$, the argument $\rightarrow -\pi/2$, so $f(x) \rightarrow -\infty$. As $x \rightarrow 1^-$, the argument $\rightarrow \pi/2$, so $f(x) \rightarrow +\infty$. Every real number is achieved. ✓

Step 3. f is continuous. For $x \in (0, 1)$ the argument $\pi(x - \frac{1}{2}) \in (-\pi/2, \pi/2)$, so no vertical asymptote of $\tan$ is encountered. Continuous. ✓

Step 4. Inverse.

$$f^{-1}(y) = \frac{1}{\pi} \arctan(y) + \frac{1}{2}, \quad y \in \mathbb{R}.$$

Since $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$, we get $f^{-1} : \mathbb{R} \rightarrow (0, 1)$. ✓

Step 5. Verify: $f^{-1}(f(x)) = \frac{1}{\pi} \arctan(\tan(\pi(x - \frac{1}{2}))) + \frac{1}{2} = x$. ✓

Step 6. $\arctan$ is continuous, so f^{-1} is continuous. ✓

Conclusion: $(0, 1) \cong \mathbb{R}$. More generally, the open unit ball $B^n \subset \mathbb{R}^n$ is homeomorphic to all of $\mathbb{R}^n$ via

$$f(\mathbf{x}) = \frac{\mathbf{x}}{1 - \|\mathbf{x}\|}, \quad f^{-1}(\mathbf{y}) = \frac{\mathbf{y}}{1 + \|\mathbf{y}\|}.$$

Key takeaway: Boundedness is NOT a topological property.

5.5 Example 5 — Any Two Open Balls in $\mathbb{R}^n$ are Homeomorphic

[Ex] Example 5: $B(p, r_1) \cong B(q, r_2)$ in $\mathbb{R}^n$

Let $B_1 = B(p, r_1)$ and $B_2 = B(q, r_2)$ be open balls in $\mathbb{R}^n$.

Step 1. Translation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ where $\mathbf{a} = q - p$.

- T moves the centre of B_1 to q .
- $T^{-1}(\mathbf{x}) = \mathbf{x} - \mathbf{a}$ (also continuous, hence a homeomorphism). ✓

Step 2. Dilation $D : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $D(\mathbf{x}) = c\mathbf{x}$ where $c = r_2/r_1 \neq 0$.

- D scales B_1 (now centred at q) to have radius r_2 .
- $D^{-1}(\mathbf{x}) = \frac{1}{c}\mathbf{x}$ (also continuous). ✓

Step 3. Composition: $\varphi = D \circ T$ maps B_1 homeomorphically onto B_2 :

$$\varphi(\mathbf{x}) = c(\mathbf{x} + \mathbf{a}) = c\mathbf{x} + c\mathbf{a}.$$

Compositions of homeomorphisms are homeomorphisms. ✓

Conclusion: $B(p, r_1) \cong B(q, r_2)$. **Size (radius) is NOT a topological property.**

5.6 Example 6 — Cube Surface Homeomorphic to Sphere

[Ex] Example 6: Surface of cube $\cong$ surface of sphere S^2

Define:

$$C = \{(x, y, z) \in \mathbb{R}^3 : \max(|x|, |y|, |z|) = 1\} \quad (\text{unit cube surface}),$$

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\} \quad (\text{unit sphere}).$$

Intuition. Think of S^2 as a balloon inside the transparent cube. Inflate it until it touches all six faces — that deformation is the homeomorphism.

Step 1. Map $f : S^2 \rightarrow C$ (sphere to cube).

$$f(x, y, z) = \frac{(x, y, z)}{\max(|x|, |y|, |z|)}.$$

For any point on S^2 at least one coordinate is nonzero, so the denominator > 0 . After dividing, max of absolute values = 1, so the image lies on C . ✓

Step 2. Map $g : C \rightarrow S^2$ (cube to sphere, inverse of f).

$$g(x, y, z) = \frac{(x, y, z)}{\sqrt{x^2 + y^2 + z^2}}.$$

On C the point is nonzero, so the Euclidean norm > 0 . After dividing, the norm equals 1, so the image lies on S^2 . ✓

Step 3. Continuity. Both f and g are ratios of continuous functions with strictly positive denominators on their respective domains. Both are continuous. ✓

Step 4. Inverse check (geometric). f sends a sphere point to the cube along the radial ray from the origin; g sends it back to the sphere along the same ray. Formally $g \circ f = \text{id}_{S^2}$ and $f \circ g = \text{id}_C$.

Conclusion: $C \cong S^2$. Having corners is NOT a topological property.

6 Topological vs. Non-Topological Properties

A property is **topological** (or a *topological invariant*) if it is preserved by every homeomorphism.

Property	Topological?	Evidence
Connectedness	Yes	Continuous images of connected sets are connected.
Compactness	Yes	Continuous images of compact sets are compact.
Number of holes	Yes	Coffee mug $\cong$ donut (one hole each).
Being a loop vs. an arc	Yes	S^1 and $(0, 1)$ are not homeomorphic.
Size / radius	No	All open balls in $\mathbb{R}^n$ are homeomorphic.
Boundedness	No	$(0, 1) \cong \mathbb{R}$ (Example 4).
Angles / corners	No	Cube surface $\cong$ sphere (Example 6).
Being a straight line	No	A straight segment $\cong$ a jagged segment.

7 Famous Example: Coffee Mug $\cong$ Donut

[Key] The Classic Example

A coffee mug and a donut (torus) are homeomorphic because both have **exactly one hole**. To see this:

1. Start with the mug (made of imaginary clay).
2. Gradually shrink and flatten the cylindrical cup part.
3. The handle forms the hole of the torus.
4. The flattened cup fills out the body of the donut.

No tearing or gluing occurs at any stage, so the map is bicontinuous.

8 Practice Problems

Q1. Linear Homeomorphism. Show that $f(x) = 5x - 2$ is a homeomorphism from $\mathbb{R}$ to $\mathbb{R}$. Find f^{-1} explicitly and verify that both f and f^{-1} are continuous.

Q2. Finite Spaces. Let $X = \{p, q, r\}$ with $T_1 = \{\emptyset, \{p\}, \{p, q\}, X\}$ and $Y = \{1, 2, 3\}$ with $T_2 = \{\emptyset, \{1\}, \{1, 2\}, Y\}$. Define $g(p) = 1$, $g(q) = 2$, $g(r) = 3$. Use condition $(\star)$ to determine whether g is a homeomorphism.

Q3. Open Intervals. Find an explicit homeomorphism $\varphi : (-1, 1) \rightarrow (3, 7)$. Write down both φ and φ^{-1} , and verify they are inverse to each other.

Q4. Bijective but Not a Homeomorphism. Consider $f : [0, 2\pi) \rightarrow S^1$ defined by $f(t) = (\cos t, \sin t)$. This map is bijective and continuous.

- (a) Show that f is continuous.
- (b) Show that f^{-1} is *not* continuous by considering a sequence of points on S^1 near $(1, 0)$ whose pre-images do not converge to $f^{-1}(1, 0) = 0$.
- (c) Conclude that f is *not* a homeomorphism.

Q5. Topological Properties.

- (a) Explain why $(0, 1)$ and $[0, 1]$ are *not* homeomorphic. (*Hint:* consider what happens when you remove an endpoint from each space and think about connectedness.)
- (b) Are the letters **C** and **I** homeomorphic (viewed as 1-dimensional curves)? What about **O** and **P**? Justify your answers using the number of branch points or loops.

9 Summary

[*] Lecture Summary

- A **homeomorphism** $\varphi : X \rightarrow Y$ is a *bijective, continuous* map whose inverse φ^{-1} is also *continuous* (bicontinuous bijection).
- The **open-set criterion** ($\star$): A bijection $f : X_1 \rightarrow X_2$ is a homeomorphism $\iff U \in T_1 \iff f(U) \in T_2$.
- Homeomorphic spaces have *identical topological structure*: their topologies are just re-labellings of each other.
- Homeomorphism preserves: connectedness, compactness, number of holes, loops.
- Homeomorphism does *not* preserve: size, boundedness, angles, corners, straightness.
- **Key examples:**
 - All open balls in $\mathbb{R}^n$ are homeomorphic (size is not topological).
 - $B^n \cong \mathbb{R}^n$ (boundedness is not topological).
 - Cube surface $\cong$ sphere S^2 (corners are not topological).
 - Coffee mug $\cong$ donut (same number of holes).

Reference: Munkres, *Topology*, 2nd ed., §18 (continuous functions) & §26. Sutherland, *Introduction to Metric and Topological Spaces*, Ch. 8.